

Mathematics is the Queen that rules the Universe

Math 5750/6880 Game Theory - Spring 2026

MW 3:00 am - 4:20 pm AEB 306

Instructor: Professor Elena Cherkhev

Email: elena@math.utah.edu

Office hours: W 10:30-11:30 am LCB 206 and by appointment

Please include '5750' in the subject line when sending an email.

Text: E.N. Barron, Game theory: Introduction

The textbook can be read online or downloaded from the Marriott library:
<https://ebookcentral.proquest.com/lib/utah/detail.action?docID=1157719>

Additional course materials:

T.S. Ferguson, A Course in Game Theory, 2020

M.J. Osborne, An Introduction to Game Theory, 2002

H. Kuhn, Lectures on the Theory of Games, 2003

G. Owen, Game theory, 1982

The homework will be assigned and expected to be done.

Exams: There will be two midterm exams, a final, and an optional project.

The grade is an average of the scores earned.

Tentative dates for the midterms: Feb 25; April 6.

Final test: Monday, April 27, 2026, 3:30 – 5:30 pm

Last day to add without a permission code: Friday, January 9

Last day to add, drop, elect CR/NC, or audit classes: Friday, January 16

Last day to withdraw from classes: Friday, March 6

Classes end: Tuesday, April 21

Holidays

Martin Luther King Jr. Day: Monday, January 19

Presidents Day: Monday, February 16

Spring Break: March 7-15

Content overview Conflicts are absolutely unavoidable in every aspect of our lives, and their rational resolution and making an optimal decision is critical in every action when one needs to make a choice. A mathematical theory of games is a theory of modeling conflicts between different players (or parties, or agents) and methods of constructing optimal solutions (or making good decisions). Applications are numerous; we will discuss mathematical formulations and solutions to games or conflict problems in economics, politics, computer science, and social and behavioral sciences.

The course gives an overview of ideas, models, and methods in mathematical game theory that studies conflicts and interactions among independent agents or players. The class covers two-person and n-person games and methods of constructing their solutions. We will discuss the von Neumann minimax theorem and discuss various methods of finding saddle point solutions in pure and mixed strategies. We will discuss the Linear Programming (LP) method, which solves a large class of optimization problems with linear equality and inequality constraints and is one of the powerful methods of solution of matrix games and minimax optimization problems. We will also talk about Nash equilibrium and the NonLinear Programming (NLP) method, which can be used to find Nash equilibria. We will discuss n-person cooperative games and different ways to define 'fairness' and to determine imputations or allocations. We will discuss a wide variety of examples illustrating the theory and modeling conflict problems arising in different applications.

Course objectives and expected learning outcomes

The course gives an overview of ideas, models, and methods in mathematical game theory. Upon successful completion of the course, a student should be able to:

- Understand, utilize, and manipulate basic concepts of game theory
- Understand the theory of saddle point solution and solution in pure and mixed strategies
- Learn algorithms for finding solutions for two-person games
- Learn the Linear Programming (LP) method of solving optimization problems with linear equality and inequality constraints
- Have a basic understanding of the concept of Nash equilibrium
- Learn efficient methods of constructing solutions for matrix games
- Understand basic principles of modeling and solving cooperative games

Table 1: Tentative schedule (Could be adjusted during the semester)

No	Date	Subject	Text/pages
		Two-person zero-sum matrix games	Ch. 1
1	Jan 5 - 9	Introduction. Two-person zero-sum matrix games. Dominated strategies, pure and mixed strategies.	Ch. 1
2	Jan 12 - 16	Graphical solution. The best mixed strategies. Lower and upper values of the game. Saddle point solution in pure strategies.	13-16
3	Jan 19 - 23	The von Neumann minimax theorem - Fundamental Theorem for all matrix games.	18-20
4	Jan 26 - Jan 30	Methods of finding saddle point solution in mixed strategies for two-person zero-sum matrix games.	25-43
5	Feb 2 - 6	Graphical method of solution. The best response strategy (the strategy of the opponent is known).	45-48
		Solution methods for matrix games	Ch. 2
6	Feb 9 - 13	Solution methods for invertible matrix games. Completely mixed games. Symmetric games.	60-79
7	Feb 16 - 20	Matrix games and Linear Programming (LP) method	82-101
		Two-person nonzero sum games	Ch.3
8	Feb 23 - Feb 27	Nash equilibrium in nonzero sum games. Individual rationality, guaranteed payoffs, rational reaction sets.	118-129
9	Mar 2 - 6	Nash equilibrium in bimatrix games and methods of its computation. Rational reaction sets. Generalizations to n -dimensional case.	128-143
**	Mar 7 - 15	*** Spring Break ***	***
10	Mar 16 - 20	Nash equilibrium strategies in completely mixed nonzero-sum games. NonLinear Programming (NLP) method.	148-156
		N-person nonzero sum games and games with a continuum of strategies	Ch. 5
11	Mar 23 - 27	N -person nonzero sum games with a continuum of strategies.	Ch. 5
12	Mar 30 - Apr 3	Mixed strategies in a continuum setting. Classes of games with continua of strategies. Timing Games.	
		N-person cooperative games	Ch. 6
13	Apr 6 - 10	A characteristic function, imputations, allocations. Domination. The core of undominated allocations.	288-310
14	Apr 13 - 17	The Shapley value. Voting systems.	335-347
15	Apr 20	Reserved	

The syllabus may be changed The syllabus is meant to serve as an outline and guide for our course. Please note that I may modify it with reasonable notice to you. I may also modify the course schedule and dates of the midterms depending on the needs of our class. Any changes will be announced in class and posted on Canvas.

Plagiarism is unacceptable and results in an **F** grade.

Academic Code of Conduct: Students are encouraged to review the Student Code for the University of Utah:

<https://regulations.utah.edu/academics/6-410.php>.

In order to ensure that the highest standards of academic conduct are promoted and supported at the University, students must adhere to generally accepted standards of academic honesty, including refraining from cheating, plagiarizing, research misconduct, misrepresenting one's work, and/or inappropriately collaborating.

A student who engages in academic misconduct may be subject to academic sanctions including but not limited to a grade reduction and failing grade.

The Office of the Dean of Students is dedicated to being a resource to students through support, advocacy, involvement, and accountability. Please consider reaching out to the Office of Dean of Students for any questions, issues and concerns. <https://deanofstudents.utah.edu>.

The American with Disabilities Act requires that reasonable accommodations be provided for students with physical, systemic, learning, and other disabilities. The University of Utah seeks to provide equal access to its programs, services and activities for people with disabilities. Please contact me at the beginning of the semester to discuss any such accommodations for the course.